

NTWALI HERVÉ

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EDUCATION

Ajou University

Bachelor of Software Engineering (4th Year, Senior)

Suwon, South Korea

March 2023 — Feb 2027

- Cumulative GPA: 3.2/4.5 (87/100) | Recipient of the **Global Korea Scholarship (GKS)** full government grant.
- Relevant Coursework:** Systems Programming, Operating Systems, Algorithms & Data Structures, Machine Learning, Database Systems, Computer Networks, Linear Algebra, Probability & Statistics.

Daegu University

Korean Language Immersion Program (GKS)

Gyeongbuk, South Korea

March 2022 — Feb 2023

- Completed intensive one-year academic Korean language training from foundational to advanced levels.
- Attained **TOPIK Level 5 (Advanced)**, demonstrating fluent proficiency for technical collaboration and research.

OPEN SOURCE & WORK EXPERIENCE

Active Member & Contributor

FOSSFORALL Korea (fossforall.org)

July 2026 — Present

Seoul, South Korea

- Actively participating in open-source community initiatives advancing Linux desktop tooling, Free & Open Source Software (FOSS), and open science.
- Collaborating on open utility development, shell scripting automation, and bilingual technical documentation.

Open Source Contributor

Quartz ([FOSS Project page](#))

March 2024 — Present

Remote / GitHub

- Contributed to an open-source static site engine built on TypeScript and Preact for publishing connected knowledge bases.
- Customized AST markdown parsing workflows and developed responsive SCSS layout components for interactive knowledge graph visualizations.
- Configured automated CI/CD build and deployment pipelines using GitHub Actions and Vercel.

ENGINEERING PROJECTS

Developer, Aidoku WebAssembly Source Engine (github.com/punishedzoomer/ocecomic)

Jan 2026 — Present

- Engineered high-performance `#![no_std]` WebAssembly (Wasm) content extraction modules in Rust for memory-constrained mobile environments.
- Implemented streaming DOM traversal and resilient HTTP request pipelines utilizing `aidoku-rs` abstractions.
- Optimized binary artifacts via link-time optimization (LTO) and dead code stripping to ensure near-zero memory footprint.

Developer, Raychess — C Chess Engine & Rendering Core (github.com/punishedzoomer/raychess)

Nov 2025 — June 2026

- Built a standalone real-time chess engine entirely in C with Raylib for GPU-accelerated rendering.
- Implemented bitboard state representations, move-generation logic, castling, and en passant validation algorithms.
- Handled manual heap/stack allocations and optimized pointer structures for deterministic, zero-overhead execution on Unix/POSIX systems.

EXTRACURRICULAR COMPETITIONS

National Cyber League (NCL) CTF Competition

Oct 2020 — Jan 2021

- #11 Nationally out of 957 collegiate teams (Top 1%)** in the NCL Team Game (Gold Bracket).
- Placed in the **99th percentile** in Password Cracking and **90th percentile** in Cryptography during the Individual Game.
- Conducted high-volume Linux server log forensics, network traffic packet analysis (PCAP), and vulnerability assessments.

CERTIFICATIONS & TRAINING

Google AI Studio: Build and Deploy, Coursera / Google Certification (<https://coursera.org>)

May 2026

- Professional certification covering agentic workflow design, API integration, and automated prototyping pipelines.

AWS Cloud Practitioner Essentials, Amazon Web Services (AWS) (<https://aws.amazon.com>)

May 2026

- Comprehensive training in cloud infrastructure, Linux compute instances, networking, IAM security, and server reliability.

TECHNICAL SKILLS

- Systems & Programming:** C, C++, Rust (Wasm), Python, Bash/Shell Scripting, TypeScript, JavaScript, SQL.
- Infrastructure & Tools:** Linux / Unix CLI, Nginx Reverse Proxies, Git / GitHub Actions (CI/CD), Docker, Vercel.
- Human Languages:** English (Fluent / Native Working), Korean (**TOPIK Level 5** — Advanced), French (Native / Fluent), Kinyarwanda (Native).